Varun Sahni *LLM Systems, Multi-Agent GenAI, On-Device AI*

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Education

M.S., Computer Science, *New York University, Courant Institute of Mathematical Sciences* — Aug. 2026 – May 2028 (Expected)
Focus: Machine Learning, Large Language Models, AI Systems.

B.E., Computer Science, *BITS-Pilani, Pilani Campus* — 2020 – 2024
Coursework: Data Structures & Algorithms, Operating Systems, Compilers, Computer Networks, Databases, Computer Architecture. — GPA: **8.11/10**

Experience

Software Engineer, GenAI / LLM Systems — July 2024 – Present
Qualcomm — *Hyderabad, India*

- Architected **Code Buddy**, a multi-agent GenAI code-review platform deployed across **4,000+** repos for **12,000+** engineers, running detect-then-fix agents under a tool-calling orchestrator that generate self-validating fixes and clear a static-analysis guardrail before posting.
- Scaled the platform to **2M+** automated code reviews (**180,000+** monthly), saving an estimated **114,000+** engineer-hours per month.
- Cut review-retrieval latency via a Tree-sitter symbolic-retrieval layer over a ccls/ctags dependency graph with TTL symbol caching.
- Benchmarked agent precision/recall on golden datasets with prompt-evaluation harnesses, enabling safe iteration across model upgrades.
- Automated interactive 4G/5G NAS protocol testing with an agentic tool that reasons over C/C++ modem source through a **14-tool** calling loop.
- Built an autonomous LLM agent that triages NAS modem hotline escalations end-to-end via a **13-tool** calling loop: it parses issue emails, searches Perforce source and CR/JIRA databases, and shelves a 3GPP-cited candidate fix after a **4-model** consensus review.

Software Development Intern, Performance / Systems — Jan. 2024 – Jun. 2024
Advanced Micro Devices (AMD) — *Bangalore, India*

- Cut bus-trace analysis runtime from **a week to 45 minutes** by building a multi-threaded LSF pipeline that streamed and sharded **250 GB+** datasets.
- Automated triage of **1,000+** JIRA issues via the REST API with hierarchical linkages and auto-summaries.
- Drove hardware-selection decisions by benchmarking CPUs and servers with Sysbench and SPEC CPU 2017 on throughput and tail latency.
- Authored reusable benchmark suites and automation scripts that standardized the team's performance-analysis workflow across platforms.

Software Development Intern — May 2023 – Jul. 2023
Standard Chartered GBS Pvt Ltd — *Chennai, India*

- Hardened the **SC Pay** backend with **100+** JUnit and Mockito tests, achieving **98.61%** line coverage.
- Migrated **10+** repositories from Jenkins to Azure DevOps with CI/CD; earned the **CCIB SCPay 2023 Award**.

Selected Projects

CabinAI, Hybrid Multi-Agent VLA for Automotive In-Cabin Intelligence — 2026
MediaPipe, Ollama (Qwen-7B INT4), Qwen3-VL-32B, gpt-oss-20b, Qualcomm AIC100, BGE-small ONNX, ChromaDB

- Architected a **7-agent hybrid edge-cloud vision-language-action system** for real-time driver monitoring: 30 FPS MediaPipe face-mesh DMS and Qwen-7B INT4 (**40 tok/s**) handle safety and voice on-device, escalating complex reasoning to Qwen3-VL-32B / gpt-oss-20b on Qualcomm AIC100.
- Routed **80%+** of queries to edge via a sub-1 ms rule-based router, with multi-tier fallback (AIC100, QGenie Claude, mock) keeping every agent online.
- Built a **15-minute-ahead** proactive fatigue forecast at **500 ms** by sending a **13.5 KB** INT8 buffer plus the latest cabin frame to a Qwen3-VL-32B VLM.
- Powered local RAG over a 31-document corpus in <**50 ms** via BGE-small ONNX (no PyTorch); preempts non-safety bus events when drowsiness > 0.7.

MyAI, Privacy-First Personal On-Device AI Assistant — 2026
Kotlin, Android (SDK 35), Qualcomm Genie SDK + QNN v2.45, Hexagon NPU v79, Qwen2.5-1.5B / Llama-3.2-3B INT4

- Designed a **4-tier on-device inference router** (trie, cache, tools, INT4 LLM) that resolves most queries without loading the full LLM.
- Ran Qwen2.5-1.5B / Llama-3.2-3B INT4 at ∼**30 tok/s** (w4a16) on Hexagon NPU v79 via the Qualcomm Genie SDK and QNN runtime.
- Enforced **kernel-level privacy** by omitting the INTERNET permission, gated by a Gradle CI task (assertNoInternet) proving zero network egress.
- Exposed an on-device agent / tool layer (calculator, datetime, calendar, contacts, alarm) that the LLM invokes via structured JSON tool-calls.
- Added a feedback-driven learning cache that, after one thumbs-up, serves a repeat query in <**10 ms** instead of ∼**600 ms** of model inference.

HealthSense, On-Wearable Multimodal Biosignal Foundation Model — 2026
PyTorch, ONNX Runtime + QNN EP, Hexagon NPU, INT8 quantization, PhysioNet MIT-BIH AFib

- Trained **PanLUNA**, a **5.4M-param** multimodal Transformer foundation model that fuses five wearable biosignals (EEG, ECG, PPG, EMG, IMU) into a shared 1024-d latent, then routes it to 4 task heads detecting cardiac arrhythmia, sleep stage, stress, and EEG abnormality from one forward pass.
- Ran INT8 inference on Hexagon NPU at ∼**95 ms / 19 mJ per pass** with a one-line ONNX hot-swap between real and synthetic cardiac models.
- Validated the cardiac head on **PhysioNet MIT-BIH AFib** (530 windows, 4 records): **96.8%** AFib sensitivity, **91.7%** sinus specificity.
- Built **BioTrain**, an on-device fine-tuning loop that personalises the model to each wearer (last-2-block backprop + GroupNorm in 0.67 MB RAM), lifting stress-detection F1 from **0.78 to 0.91** in ∼**3.6 s** under **50 mW** while keeping all raw biosignals on the device.

Compiler Construction (C) — Feb. 2023 – Apr. 2023
Academic Project — *BITS Pilani*

- Built an end-to-end compiler (DFA twin-buffer lexer, LL(1) parser with FIRST/FOLLOW gen, AST, codegen) cutting parse-tree memory by **70%**.
- Added semantic analysis with type checking and scope resolution, plus panic-mode error recovery for graceful handling of malformed input.

Achievements

- **Won SparQ Qualcomm TechFest 2026**: presented the CabinAI architecture to the VP of AI and VP of Automotive at Qualcomm. — 2026
- **SparQ Qualcomm TechFest 2025 Finalist**: presented the Code Buddy multi-agent architecture to the Qualcomm CTO. — 2025
- **DAAD Scholarship**, German Academic Exchange Service: selected for Mixed Reality research at TU Braunschweig, Germany. — 2022
- **CCIB SCPay 2023 Award**, Standard Chartered Bank, for the Jenkins to Azure DevOps migration. — 2023
- **JumpStart Hackathon**, Publicis Sapient: top performance, earned exclusive program access. — 2023
- **JEE Advanced**: All-India Rank **2358 / 1.2M+** (top 0.2%), Indian Institutes of Technology. — 2020

Leadership

- **Founding President, Kalipatnapu AR/VR Lab, BITS Pilani** (2023 – 2024): led a 40-student team and ran a **Rs. 40,000+** VR experience zone.
- **Blockchain Dept Coordinator, BITS Coding Club** (2022 – 2023): chartered the new department, ran recruitments, organized an **ICP Hackathon**.

Technical Skills

Languages: C, C++, Python, Java, Kotlin, JavaScript, SQL, Bash, C#.
ML / LLM Stack: PyTorch, Hugging Face Transformers, LangChain, vLLM, Ollama, ONNX Runtime, RAG, LoRA / QLoRA fine-tuning, multi-agent orchestration, prompt evaluation, structured-output / tool use, embeddings (Tree-sitter, BGE, ChromaDB).
Edge / On-Device AI & Systems: Snapdragon AIC100, Hexagon NPU, Qualcomm Genie SDK + QNN, AI Hub, INT4/INT8 quantization, Llama.cpp; Linux, Docker, Kubernetes, Redis, Flask, Azure DevOps, Git, CI/CD.